

KBTG: Innovating financial management for young Thais with an innovative, scalable mobile app

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ABOUT KASIKORN



About KASIKORN Business - Technology Group (KBTG)

With Google Cloud, KASIKORN Business Technology Group has applied innovation, experimentation, budgeting, mobile, and meets the needs of Thais.

KASIKORN Business - Technology Group (KBTG) operates as the IT enabler of KBank, one of the largest banks in Thailand. The Group provides the infrastructure and innovation that enables KBank to operate efficiently, deliver high quality customer experiences, and stay ahead of the innovation curve.

Google Cloud results

- Exceeded many traditional banking applications with average user rating of 4.8/5 on leading application store
- Recorded no downtime attributable to cloud infrastructure
- Shortened software development cycle from months to weeks

Scale
mobile
seamless
millions
users
500,000

Industries: Financial Services & Insurance

Location: Thailand

(<https://cloud.google.com/anthos/service-mesh/>)

Firebase (<https://firebase.google.com/>)

Firebase Cloud Messaging

(<https://firebase.google.com/products/cloud-messaging/>)

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)

Service Mesh

(<https://cloud.google.com/anthos/service-mesh>)

Google Cloud Deploy

(<https://cloud.google.com/deploy>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine

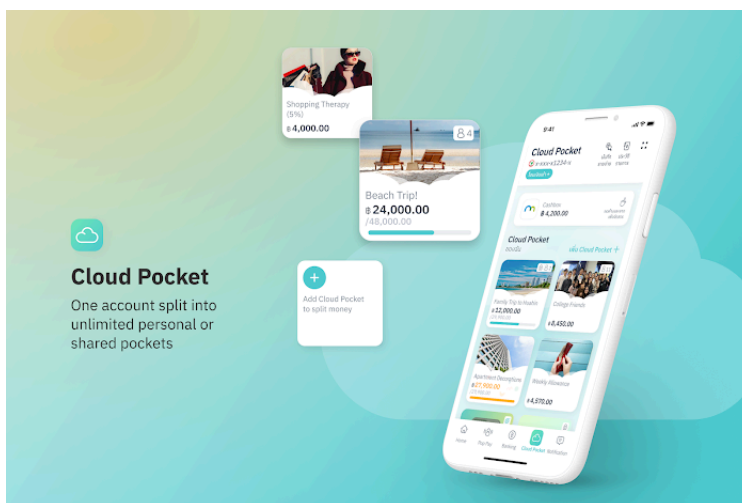
(<https://cloud.google.com/kubernetes-engine/>)

The banking and financial services sector is transforming rapidly to meet changing customer demands and seize opportunities to expand into new markets and user segments. KASIKORNBANK (known as KBank), one of the largest financial institutions in Thailand, is turning to technology leaders, such as Google Cloud

(<https://cloud.google.com/>), to help execute a \$2.7 billion plan to become a regional digital bank servicing South East Asia, including new demographic groups in its home market.

KBank has assigned the technology component of its transformation to subsidiary KASIKORN Business - Technology Group (KBTG). KBTG has two complementary responsibilities, to provide IT infrastructure and services that enable KBank to run its banking and financial services, including mobile banking applications and branch software, and pursue technology-driven innovation and the future of finance.

"We envision how banking and finance will remain relevant to consumers over the next decade and create new technologies and solutions that propel KBank into that future," says Thadpong Pongthawornkamol, Managing Director, KBTG. "This means developing new ways of serving customers and running our back office operation."



KBTG's MAKE app's Cloud Pocket feature demonstrated on a smartphone

MAKE application as one of KBTG Labs' biggest success stories

KBTG established KBTG Labs to deliver its innovation activities. The KBTG Labs team comprises business, product, and technology specialists who collaborate to create the innovation prototypes that KBTG tests and deploys in the market. A mobile app, MAKE, that enables younger Thai people to manage their financial circumstances is one of KBTG Labs' biggest success stories, reaching one million users in three years. The business elected to develop and deploy MAKE on Google Cloud.

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Managing Director, KBTG

KBank developed the concept of MAKE in 2019 to compete with emerging digital-native banking apps and to complement its incumbent full-featured K PLUS mobile application that has 20 million users. "We were running K PLUS at that time in an on-premises infrastructure, but the maturity of cloud in 2019 meant it made sense to develop and run MAKE in a cloud-native technology stack," says Pongthawornkamol. "With Google Cloud, we could look beyond optimizing technology engineering, architecture and maintenance to executing our broader ambition of rebuilding digital banking."



Support for innovation and experimentation drives cloud choice

KBTG and KBTG Labs had undertaken an extensive review of cloud options and based on criteria, including ease of development, innovation and experimentation, and scalability, determined Google Cloud was the best fit for MAKE. "Our usual approach is to apply scoring and other formal processes and techniques to ensure rigor in the selection of technologies that support and enable our work with KBank," explains Pongthawornkamol. "However, when we set up MAKE, we put to the KBank Technical Advisory Council that we wanted to create an exception in this case and select Google Cloud for its capacity to support experimentation through to implementation at scale."

"Google Cloud enables a technology and innovation playground with MAKE," says Pongthawornkamol. Creative steps include using the Flutter open source framework developed by Google to build a platform-agnostic MAKE application front-end for deployment at scale. "Previously, when developing applications across iOS and Android, we would have to manually investigate and resolve production issues that occurred due to differences in implementation processes between the two mobile operating systems," explains Pongthawornkamol. "With Flutter, we do not experience these problems, which

expedites the development process and frees our developers to focus on more important activities."

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KBTG uses Firestore Cloud Messaging

(<https://firebase.google.com/docs/cloud-messaging>) to send cross-platform notifications to MAKE users and Firebase A/B Testing

(<https://firebase.google.com/docs/ab-testing>) to segment users by demographic, device hardware, device operating system, or other relevant factors to test planned features or updates. "With Firebase A/B Testing, we can quickly determine whether a new feature will resonate with targeted groups and identify where we need to tweak or restructure, in line with our fast-paced approach to MAKE development," explains Pongthawornkamol.

Shortened development cycles enable rapid response

KBTG employs a range of Google Cloud products and services to enable its team to focus on experimentation and innovation for MAKE, rather than on infrastructure management tasks. These products include Service Mesh

(<https://cloud.google.com/anthos/service-mesh>) that automates and manages communications between microservices, and Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) that orchestrates resources across the containerized microservices environment, enabling KBTG to accelerate its development lifecycles and scale without disruption. "With Google Kubernetes Engine and other Google Cloud services, we have grown seamlessly to 500,000 monthly average users since we launched MAKE," says Pongthawornkamol.

"We've shortened development cycles from months to weeks, meaning we can respond quickly to new

opportunities and feedback in creating new features for the application."

Building and running MAKE in Google Cloud has enabled KBTG to build knowledge and foster experimentation across the organization in areas, such as database services, DevOps, and continuous integration/continuous development (CI/CD). After initially running, the MAKE database used to manage financial and customer data in Google Kubernetes Engine, the business then elected to move it to Cloud SQL (<https://cloud.google.com/sql>), so while Cloud Build (<https://cloud.google.com/build>) provides a serverless CI/CD platform that enhances the development pipeline, Cloud Deploy (<https://cloud.google.com/deploy>) enables continuous delivery to Google Kubernetes Engine.

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The business has experienced only minor outages during the development and operation of MAKE, with none caused by its cloud infrastructure. This high availability combined with KBTG's innovative, user-centric approach to developing MAKE is reflected in the positive user response to the new mobile application. As at late January 2023, the application had an average score of 4.8 from 44,000-plus reviews on Google Play, well above the average review score of some more established mobile and digital banking applications.

"With Google Cloud, we've created a vibrant, relevant budget management application for younger Thais that complements our existing established digital banking property," explains Pongthawornkamol.

"We've implemented best practices in software development and cloud architecture, and will be pushing these across KBTG and KBank. Our next step with Google Cloud will be to deploy machine learning and data analytics to provide an even more relevant, meaningful experience to users of MAKE."

